

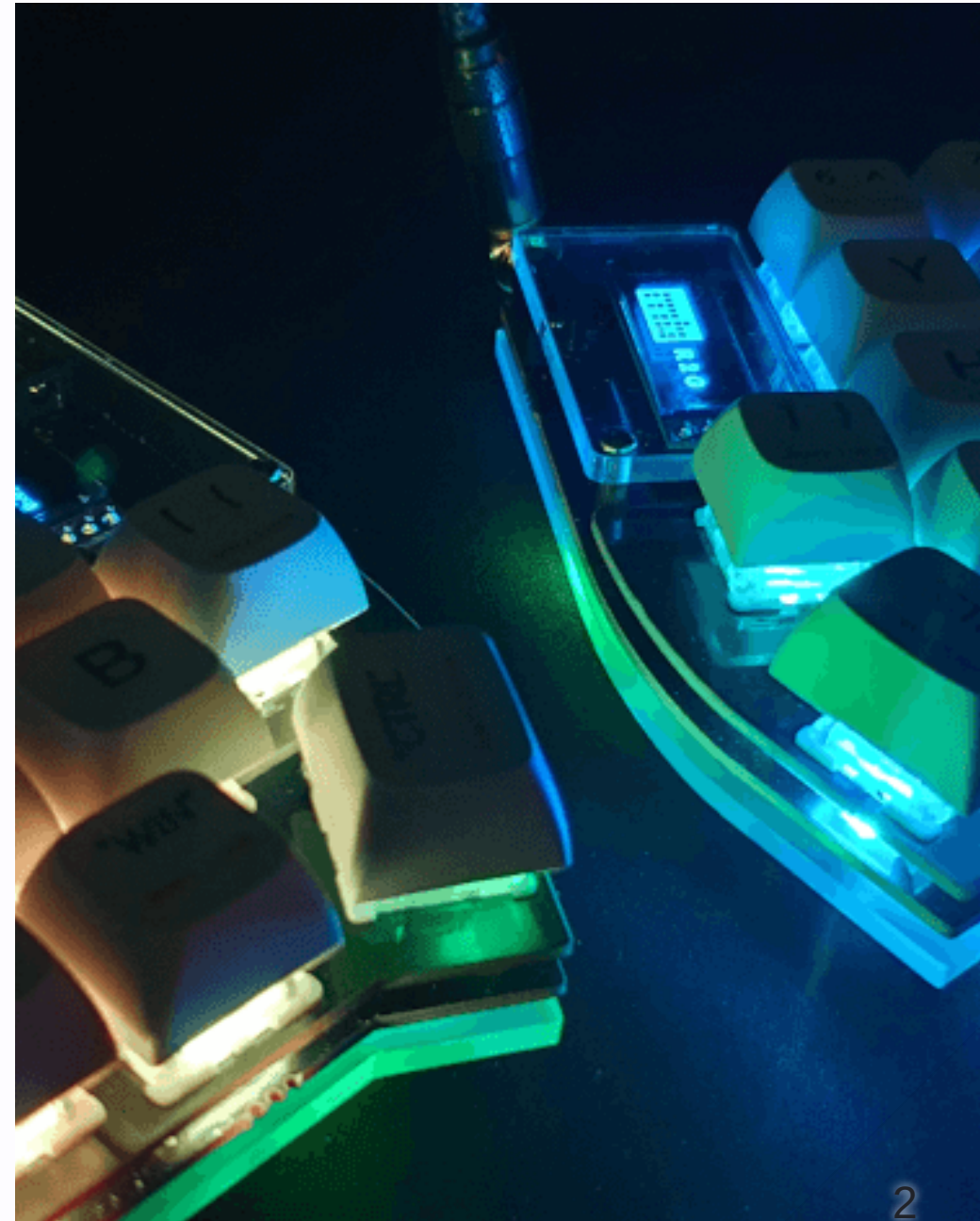
Using HTMX To Make Interactive Elements In Drupal

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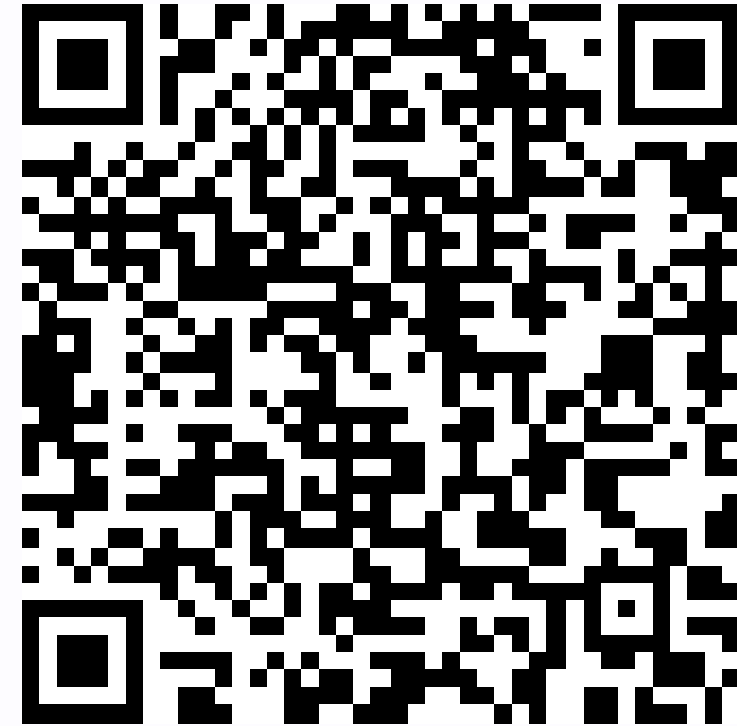
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Source Code

- Talk is available at:
<https://github.com/hashbangcode/drupal-hmx-talk> or via QR code
- All code seen can be found at:
 - <https://bit.ly/46arnzj> HTMX examples
 - <https://bit.ly/3OHAIIR> Drupal examples
- More resources at
www.hashbangcode.com



Using HTMX To Make Interactive Elements In Drupal

HTMX In Drupal

- HTMX added to Drupal in 11.3 as a core component.
- This is the new standard for ajax requests.
- All existing ajax features will be re-written in HTMX.

HTMX

What is HTMX?

- JavaScript framework.
- Allows ajax calls and CSS Transitions *without writing JavaScript code*.
- Small at 16K (minified and gzipped).
- Uses HTML attributes.
- Plugins extend functionality.
- Platform agnostic.
- No other dependencies.

What is HTMX?

- Behaviour is added to HTML attributes.
- Any element can issue a web request.
- All responses should be in HTML.

Installing HTMX

- Download and include the single JavaScript file.

```
<script src="htmx.min.js"></script>
```

- There are CDN options available.
- We are now ready to use HTMX.

HTMX Attributes

- Prefixed with `hx-` .
- 11 attributes are commonly used.
- There are another 20-30 that add more functionality.

HTMX Attributes - Example

```
<button hx-get="index.php" hx-target="#div1">Submit</button>  
  
<div id="div1"></div>
```

This example will:

- `hx-get` = Send a "get" request to the index.php path.
- `hx-target` = Write the response in the element with the ID "div1".

HTMX Requests

- All requests from HTMX will have the `hx-request` header.
- Certain attributes will add further headers to the request.

Responding To The Request

Your response should be in plain HTML.

We could respond to the request with:

```
<p>Button clicked!</p>
```

Which would be injected into our page like this.

```
<div id="div1"><p>Button clicked!</p></div>
```

Responder Class In PHP

A simple class to respond to HTMX requests.

```
class Htmx {  
    public static function isHtmxRequest(): bool {  
        return isset($_SERVER['HTTP_HX_REQUEST']) && $_SERVER['HTTP_HX_REQUEST'] === 'true';  
    }  
  
    public static function isGet(): bool {  
        return $_SERVER['REQUEST_METHOD'] === 'GET';  
    }  
}
```

Responding To The Request

We detect the `hx-request` header and the correct HTTP method before responding.

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo '<p>Button clicked at ' . date('r') . ' .</p>';  
}
```

The div now looks like this:

```
<div id="div1"><p>Button clicked at Sun, 08 Feb 2026 16:00:23 +0000.</p></div>
```

HTMX Common Attributes

HTTP Verbs

Different HTTP verbs are available through attributes.

- `hx-get` does a "get" request.
- `hx-post` does a "post" request.
- `hx-delete` does a "delete" request
- `hx-patch` does a "patch" request
- `hx-put` does a "put" request

hx-target

- The target element to be swapped.

```
<button hx-get="index.php" hx-target="#div1">Submit</button>  
  
<div id="div1"></div>
```

hx-trigger

- Defines the action that will trigger the request.

```
<div hx-post="index.php" hx-trigger="click">Click me</div>
```

- Can be left out for elements that have default triggers.
- Eg. `button` elements automatically have a `click` trigger.

hx-trigger Examples

```
<div hx-post="index.php" hx-trigger="click">Click me</div>
```

```
<div hx-post="index.php" hx-trigger="click once">Click me</div>
```

```
<div hx-post="index.php" hx-trigger="revealed"></div>
```

```
<div hx-post="index.php" hx-trigger="load delay:500ms"></div>
```

```
<input hx-get="index.php" hx-trigger="changed delay:1s" />
```

hx-select

- Select content to swap with the target element from the response.

HTMX:

```
<button hx-get="index.php" hx-select="#response" hx-swap="outerHTML">Click</button>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo '<p id="response">Button clicked at ' . date('r') . ' .</p>';  
    echo '<p>Some extra content that won\'t get displayed.</p>';  
}
```

hx-select-oob

- Select "Out Of Band"
- Do more than one thing in the request.

HTMX:

```
<button hx-get="index.php" hx-select-oob="#div1">Send Request</button>  
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo 'Request Sent';  
    echo '<div id="div1">Button clicked at ' . date('r') . ' .</div>';  
}
```

hx-select-oob - Continued

- You can comma separate this property to do multiple things.

HTMX:

```
<button hx-get="index.php" hx-select-oob="#div1,#div2">Send Request</button>  
<div id="div1"></div> <div id="div2"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo 'Request Sent';  
    echo '<div id="div1">Button clicked at:</div>';  
    echo '<div id="div2">' . date('r') . '</div>';  
}
```

hx-swap

- Controls how an element will be swapped into the page.
- Defaults to `innerHTML` - replace the HTML contents.
 - `outerHTML` - Replace the element.
 - `textContent` - Replace contents without parsing as HTML.

hx-swap - Continued

- `beforebegin` , `afterbegin` , `beforeend` , `afterend` - Place the response before or after the element and its children.
- `delete` - Delete the target element from the page.
- `none` - Do nothing with the response (but still process out of band items).

hx-swap-oob

- Swap Out Of Band
- Similar to `hx-select-oob` but the response tells HTMX what to swap.

HTMX:

```
<button hx-get="index.php" hx-swap="none">Send Request</button>  
<div id="div1"></div>
```

Response:

```
if (Htmx::isHtmxRequest() && Htmx::isGet()) {  
    echo '<div id="div1" hx-swap-oob="true">Button clicked at ' . date('r') . '</div>';  
}
```

CSS Transitions

- HTMX adds classes to elements during different events.
- `htmx-added` is added to new content before it is added to the page.
- We can use these classes to animate what HTMX is doing with CSS transitions.

CSS Transitions

The following will fade in a bit of content inside the `div1` element as it is loaded into the page.

```
#div1 p {  
    opacity: 1;  
    transition: opacity 1s ease-in;  
}  
#div1 p.htmx-added {  
    opacity: 0;  
}
```

Configuring HTMX

- It is possible to configure HTMX via a number of settings.
- The preferred way to do with is via a metatag.

```
<meta name="htmx-config" content='{ "defaultSwapStyle": "outerHTML" }'>
```

Configuring HTMX

- Configuration options can be set through meta tags

```
<meta name="htmx-config" content='{ "defaultSwapStyle": "outerHTML" }'>
```

Or JavaScript.

```
<script>  
htmx.config.defaultSwapStyle = 'outerHTML';  
</script>
```

DEMO!

HTMX In Drupal

HTMX In Drupal

- HTMX is included in Drupal as a library.
- A few classes exist to assist with adding attributes and responding to requests.

HTMX in Drupal

- The standard usage of HTMX in Drupal is to use the `data-hx-` prefix for attributes.
- This is supported by HTMX.

Eg:

```
<button data-hx-get="index.php" data-hx-target="#div1">Submit</button>  
  
<div id="div1"></div>
```

Drupal Libraries

- `core/htmx` - The core HTMX library.
- `core/drupal.htmx` - Additional scripts for Drupal.
- You rarely need to inject these libraries directly.

Drupal Libraries

- `core/drupal.htmx` includes three files:
- `htmx-assets.js` - Adds assets the current page requires.
- `htmx-behaviors.js` - Connect `Drupal.behaviors` to HTMX inserted content.
- `htmx-utils.js` - Helper functions for the other two files.

Drupal Integration

- The `\Drupal\Core\Htmx` class is used as a wrapper around the HTMX libraries and attribute injection for render arrays.
It can also inject headers for responding to HTMX.
- `\Drupal\Core\Htmx\HtmxRequestInfoTrait` is used to detect HTMX requests and associated headers.

- Has a number of methods that set up attributes.

```
$output['element'] = [  
  '#type' => 'html_tag',  
  '#tag' => 'p',  
  '#value' => 'Content'  
];  
  
$htmx = new Htmx();  
$htmx->get()  
  ->swap('afterend')  
  ->trigger('revealed')  
  ->applyTo($output['element']);
```

```
<p  
  data-hx-get="/route"  
  data-hx-swap="afterend"  
  data-hx-trigger="revealed">  
Content
```

Htmx Class

- The HTMX class can also be used with a different syntax.

```
(new Htmx())  
  ->get()  
  ->swap('afterend')  
  ->trigger('revealed')  
  ->applyTo($output['element']);
```

HTMX Drupal Response

- Drupal has the ability to respond to a HTMX response with HTML.
- This uses the class `Drupal\Core\Render\MainContent\HtmxRenderer`.
- Renders the render array as a little HTML document and sends this upstream.

HTMX Drupal Response

HtmxRenderer is invoked by:

- `_wrapper_format=drupal_htmx` as a query on the incoming request.
- The `_htmx_route=true` on the route replying to the request.

HTMX Trait

- `\Drupal\Core\Htmx\HtmxRequestInfoTrait` is used to assist in detecting Htmx requests.
- Inspect what headers are attached to the request and what method is being used.

HTMX With Controllers

HTMX With Controllers

- Think about your approach.
 - Do you have two actions in your controller or one action that serves normal and HTMX traffic?
- Two actions means you need to tag one route with the route option `_htmx_route: TRUE`.
- One route means you need to add some code to your controller footprint.

HTMX With Controllers

- The `HtmxRequestInfoTrait` trait needs access to the request stack service.

```
namespace Drupal\mymodule\Controller;

use Drupal\Core\Controller\ControllerBase;
use Drupal\Core\Htmx\HtmxRequestInfoTrait;

class MyController extends ControllerBase {

    use HtmxRequestInfoTrait;

    public function __construct(protected RequestStack $requestStack) {}

    public function action() {}
}
```

HTMX With Controllers

- Then, in your action:

```
public function action() {  
  
    if ($this->isHtmxRequest()) {  
        // Respond to HTMX request.  
    }  
  
    // Respond to normal controller action.  
}
```

HTMX With Forms

HTMX With Forms

- The trait `HtmxRequestInfoTrait` is part of the `FormBase` class.
- Important!
 - The HTMX response in Forms is the entire form, so you *need* add a `hx-select` (or similar) to pick out the relevant part of the response.
 - Also, forms need to be consistent. You can't just throw elements into the form markup as the elements need to exist in the form build.

HTMX With Forms

- Works in the same way as constructors.

```
(new Htmx())  
  ->post()  
  ->target('*:has(>input[name="email"])')  
  ->select('*:has(>input[name="email"])')  
  ->trigger('keyup delay:1s')  
  ->applyTo($form['email']);
```

- This decorates the form element with the HTMX elements.

DEMO!

Druapl Needs You!

- HTMX is in Drupal 11.3 as a fully featured system.
- All of the ajax subsystem now needs to be converted to HTMX.
- There's a lot of work to be done.
- <https://www.drupal.org/community-initiatives/replace-ajax-api-with-htmx>

HTMX Module

- <https://www.drupal.org/project/htmx>
- Some extra tools and examples of HTMX in a Drupal setting.
- Has a Views plugin to display Views as HTMX.
- Has a HTMX debugger that logs all HTMX events in the browser console.

Resources

- htmx.org - <https://htmx.org>
- [HTMX Labs](https://htmxlabs.com) - <https://htmxlabs.com>

Questions?

- Slides:

<https://github.com/hashbangcode>
[htmx-talk](#)



Thanks!

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